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Editor-in-Chief
Quantum Science and Technology
IOP Publishing
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Bristol BS1 6HG, UK

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**Quantum-Enhanced Agrivoltaic Digital Twin: Non-Markovian Coherence Control, NPoM Plasmonic Diagnostics, and BB84-Secured IoT Telemetry**” for consideration as a Research Article in *Quantum Science and Technology*.

Publication route

We explicitly confirm that we opt to publish this manuscript via the **standard subscription route (carrying zero Author Processing Charge, \$0 APC)**.

Summary of the work

Agrivoltaic systems co-locating photovoltaics and crops conventionally rely on passive shading, which inherently degrades agricultural yield. To overcome this limitation, we present the first quantum digital twin for agrivoltaics, integrating non-Markovian PT-HOPS/SBD dynamics, NPoM plasmonic SERS diagnostics, microclimate FAO-56 modeling, and BB84 QKD secure IoT telemetry. In this multi-domain architecture, an organic photovoltaic (OPV) layer integrated with a nanoparticle-on-mirror (NPoM) plasmonic cavity actively engineers the light spectrum to protect quantum coherence in the crop’s Fenna–Matthews–Olson (FMO) light-harvesting complex, while simultaneously enabling non-destructive single-molecule SERS diagnostics and quantum-secured field telemetry.

Our main scientific contributions are:

- **Non-Markovian coherence control and diagnostics:** Dual-band OPV filtering extends FMO excitonic coherence lifetimes by 50% in the passive fraction and transport yields by 25% at room temperature. We resolve the NPoM plasmonic transport trade-off, establishing an optimal mode volume ($V_{\text{mode}} = 1.2 \text{ nm}^3$) that preserves a global photosynthetic yield of 97.1% across the canopy while enabling single-molecule SERS diagnostics.
- **Greenhouse microclimate and techno-economics:** A 500 m² Cameroonian cooperative greenhouse model demonstrates a 28% water saving (650 m³/yr), 90 MWh/yr of electricity generation, and a 4.23-yr payback period under 30% blended-finance grant support.
- **Agro-rural cybersecurity:** To protect agricultural data sovereignty and prevent malicious manipulation of automated irrigation, field telemetry is secured via room-temperature decoy-state BB84 QKD ($N_{\text{raw}} \approx 1200$ bits). This supplies secure, low-latency (3.7 ms) edge inference for embedded ARM Cortex-M4 IoT nodes.

Why *Quantum Science and Technology*?

Quantum Science and Technology is the premier venue for cross-disciplinary advances spanning quantum photonics, quantum biology, quantum sensing, and quantum communication. By addressing the critical challenge of food-energy-water security, our work bridges angstrom-scale non-Markovian vibronic dynamics (PT-HOPS/SBD) with field-scale agrivoltaic microclimates. We demonstrate how quantum coherence engineering and quantum cryptography can transition from fundamental physics to delivering real-world sustainability and security benefits, perfectly aligning with QST’s mandate to publish transformative quantum technologies.

Prior discussions and related submissions

A companion manuscript addressing selective vibronic excitation in isolated FMO complexes is under revision at the *Journal of Physical Chemistry Letters* (Manuscript ID: jz-2026-00994t). The present manuscript extends that quantum framework to a complete agrivoltaic digital twin incorporating NPoM cavity electrodynamics, SERS diagnostics, microclimate physics, lifecycle assessment, and BB84 cybersecurity. No part of this manuscript is under consideration elsewhere.

All authors have approved the manuscript and agree with its submission to *Quantum Science and Technology*. The authors declare no competing financial or non-financial interests.

Suggested reviewers:

1. **Prof. Shaul Mukamel** — University of California, Irvine (smukamel@uci.edu) — *Non-linear optical spectroscopy, open quantum systems.*
2. **Prof. Jeremy J. Baumberg** — University of Cambridge (jjb12@cam.ac.uk) — *NPoM nanoplasmonics, sub-nanometer strong coupling, SERS.*
3. **Prof. Alán Aspuru-Guzik** — University of Toronto (aspuru@utoronto.ca) — *Quantum biology, environment-assisted transport, digital twins.*
4. **Prof. Alexandra Olaya-Castro** — University College London (a.olaya@ucl.ac.uk) — *Biomolecular quantum dynamics, vibronic coherence.*
5. **Prof. Greg D. Scholes** — Princeton University (gscholes@princeton.edu) — *Quantum coherence in light-harvesting architectures.*

Journal cover art proposal

Upon acceptance of this manuscript, we would be delighted to submit a high-resolution 3D artistic rendering illustrating the multi-scale quantum agrivoltaic digital twin—depicting sub-nanometer NPoM plasmonic cavities coupled to crop light-harvesting complexes under a spectrally engineered OPV canopy—for consideration as a front cover image for *Quantum Science and Technology*.

Sincerely,

Steve Cabrel Tegua Kouam
(On behalf of all authors)

Enclosures: Manuscript (Manuscript_QST_26-07-29.pdf), Supporting Information (SI.pdf), and Figures.